

Order of operations



When my kids were little, my wife and I would try to teach them to get dressed. Even after they learned to put on their underwear, shorts, shirt, socks and shoes they had one last thing to learn... something which, to our great amusement took a little bit longer to grasp.

Sometimes they would put on their pants and then their underwear (a bit like superman). Other times they would get their underwear on first, then socks and shoes and struggle to get their pants over their shoes.

After a little chuckle and the obligatory photo to send to the grandparents, we would say,

"Order of operations is important."

And so it is the same for mathematics.

$2 + 3 \times 4$ will give you a different answer if you add first and then multiply or multiply first and then add:

$$2 + 3 \times 4 = 6 \times 4 = 24 \text{ (incorrect)}$$

$$2 + 3 \times 4 = 2 + 12 = 14 \text{ (correct)}$$

We have a set of rules for operational order which is meant to prevent ambiguity.

Rules:

1. evaluate innermost brackets, braces, or parentheses first and work towards the outer most

$$\left\{ a \times \left[b \times c + \underbrace{(d - e)}_{\text{1st}} / f \right] \right\}^2 + g$$

2nd

3rd

4th

2. next evaluate special functions

$$2 + \sin(30^\circ)^2 = 2 + \sin^2(30^\circ) = 2 + \left(\frac{1}{2}\right)^2 = 2 + \frac{1}{4} = 2.25$$

3. next evaluate powers, roots, and inverses
(order is interchangeable between these)

$$(a^n)^{\frac{1}{m}} = (a^{\frac{1}{m}})^n$$

4. next multiplication and division
(order is interchangeable between these)

$$(a \times b) / c = a \times (b / c)$$

5. and lastly, addition and subtraction.
(order is interchangeable)

$$(a + b) - c = a + (b - c)$$

for those steps that are interchangeable it is a bit like putting your shirt on before your pants or your pants before your shirt. The intermediate appearance may differ, but the end result is the same...

unless of course you tuck your shirt
into your underwear.